World Cup Prediction: Human (Linear Dixon–Coles) vs. Gradient Descent (MLP)

Experiment harness

June 11, 2026

1 Setup

Leak-safe split by match date: TRAIN < 2024-01-01, VALIDATION 2024-01-01 to 2025-10-01 (all model and hyperparameter selection), TEST $\geq$ 2025-10-01 (end of 2025 + the 2026 World Cup; scored once). Metric of record is the Ranked Probability Score (RPS) over the ordinal home/draw/away outcome (lower is better); we also report multiclass log-loss, Brier score, and argmax accuracy. A temporal exponential decay ($\lambda = 0.001$) weights recent matches more and is applied to both models.

Experiment 1: Isolation and convergence (validation)

Model	Config	RPS	LogLoss	Brier	Acc
Base rate	-	0.2246	1.0602	0.6398	0.467
Linear (alpha=0)	full feats	0.1889	0.9549	0.5643	0.545
MLP (weights, long)	**w128 d2 do0.2 lr1e-3 — best@11/411**	**0.1874**	**0.9456**	**0.5620**	**0.551**

Table 1: Experiment 1: Isolation and convergence (validation)

Experiment 3: Linear ridge α sweep (validation)

Model	Config	RPS	LogLoss	Brier	Acc
Linear	alpha=0	0.1882	0.9520	0.5628	0.545
Linear	**alpha=10**	**0.1880**	**0.9509**	**0.5624**	**0.542**
Linear	alpha=100	0.1884	0.9520	0.5634	0.542
Linear	alpha=1000	0.1892	0.9540	0.5655	0.542
Linear	alpha=10000	0.2016	0.9936	0.5923	0.500

Table 2: Experiment 3: Linear ridge α sweep (validation)

Experiment 4: Reduced 10-feature set (validation)

Model	Config	RPS	LogLoss	Brier	Acc
Linear (alpha=10)	10 feats	0.1890	0.9573	0.5641	0.536
MLP (best cfg)	**10 feats**	**0.1888**	**0.9513**	**0.5646**	**0.548**

Table 3: Experiment 4: Reduced 10-feature set (validation)

Experiment 5: Friendlies removed from train + val (validation)

Model	Config	RPS	LogLoss	Brier	Acc
Linear (alpha=10)	no friendlies (n_tr=1933)	0.1921	0.9815	0.5820	0.524
MLP (best cfg)	**no friendlies (n_tr=1933)**	**0.1893**	**0.9697**	**0.5786**	**0.536**

Table 4: Experiment 5: Friendlies removed from train + val (validation)

Experiment 6: ELO scale / K-factor sensitivity (validation)

Model	Config	RPS	LogLoss	Brier	Acc
Linear (alpha=10)	elo div=800 k=tourn (baseline)	0.1886	0.9533	0.5637	0.545
MLP (best cfg)	elo div=800 k=tourn (baseline)	0.1866	0.9448	0.5603	0.551
Linear (alpha=10)	elo div=200 k=50	0.1885	0.9519	0.5639	0.542
MLP (best cfg)	elo div=200 k=50	0.1870	0.9463	0.5618	0.548
Linear (alpha=10)	elo div=1600 k=50	0.1875	0.9498	0.5614	0.539
MLP (best cfg)	**elo div=1600 k=50**	**0.1860**	**0.9432**	**0.5592**	**0.551**

Table 5: Experiment 6: ELO scale / K-factor sensitivity (validation)

Final held-out TEST set (end 2025 + 2026), scored once

Model	Config	RPS	LogLoss	Brier	Acc
Base rate	-	0.2234	1.0658	0.6437	0.457
Linear (alpha=10)	full feats	0.1762	0.9143	0.5449	0.556
MLP (best cfg)	**{'lr': 0.0005, 'depth': 3, 'width': 256, 'dropout': 0.0, 'grad_clip': 5.0}**	**0.1756**	**0.9170**	**0.5444**	**0.564**

Table 6: Final held-out TEST set (end 2025 + 2026), scored once

Experiment 2: MLP hyperparameter sweep (validation)

Best 12 of 55 configurations by validation RPS. Across the whole grid RPS spanned only 0.1866–0.1900, i.e. architecture had little effect.

Model	Config	RPS	LogLoss	Brier	Acc
MLP	**lr0.0005 d3 w256 do0.0**	**0.1866**	**0.9448**	**0.5603**	**0.551**
MLP (best, clip OFF)	lr0.0005 d3 w256 do0.0	0.1866	0.9448	0.5603	0.551
MLP	lr0.003 d3 w64 do0.0	0.1867	0.9436	0.5600	0.557
MLP	lr0.0005 d2 w128 do0.2	0.1869	0.9440	0.5607	0.563
MLP	lr0.0005 d2 w128 do0.0	0.1870	0.9445	0.5612	0.554
MLP	lr0.003 d2 w128 do0.0	0.1871	0.9447	0.5614	0.557
MLP	lr0.001 d3 w256 do0.2	0.1871	0.9462	0.5615	0.563
MLP	lr0.001 d2 w128 do0.0	0.1871	0.9440	0.5609	0.548
MLP	lr0.003 d2 w128 do0.2	0.1871	0.9437	0.5611	0.551
MLP	lr0.001 d3 w64 do0.2	0.1872	0.9461	0.5613	0.557
MLP	lr0.0005 d3 w128 do0.0	0.1874	0.9472	0.5624	0.554
MLP	lr0.003 d3 w64 do0.2	0.1874	0.9460	0.5615	0.557

Table 7: MLP sweep: best 12 configurations by RPS

World Cup 2026 Monte Carlo simulation

Each model's predicted goal expectations drive 10,000 simulated tournaments (group stage, best-eight third places, full knockout bracket from openfootball/worldcup.json). Matches are treated as neutral by averaging both home/away orientations. Knockout ties (penalty shootouts) are resolved by a coin flip weighted by the teams' goalkeeper ratings, $P(A) = \mathrm{gk}_A/(\mathrm{gk}_A + \mathrm{gk}_B)$.

Shootout-resolution sensitivity (GK vs. strength)

We compare resolving knockout draws by goalkeeper rating against the simpler overall-strength (λ) rule. The effect on championship odds is small, since shootouts decide only a minority of knockout games.

Methodology changelog

Every change introduced during development, with its implementation:

1. **FIFA-year alignment fix.** Matches were keyed to squad ratings a year early. Corrected the rule to `month` $\geq$ `10` $\Rightarrow$ `year+1` in `prepare_training_data`, so each match merges with the squad of the game that was live at the time.
2. **Linear feature standardization fixed.** Standardization had been recomputed per call and applied to the coefficients (nullifying the intercept). Now the train-set mean/std are fit once in `fit()` and reused at predict time; the intercept column is appended *after* scaling. Validation log-loss fell from 3.54 to 0.94.
3. **Dixon–Coles τ in the linear likelihood.** ρ was inert because the linear NLL multiplied two independent Poissons. Added the low-score τ correction for the (0,0)/(1,0)/(0,1)/(1,1) cells; ρ now fits to ≈ -0.05.

Team	Champ (Lin)	Champ (MLP)	Champ (Ens)	Final (Lin)	Final (MLP)
Germany	7.6%	22.4%	15.0%	13.3%	32.9%
Spain	14.0%	15.4%	14.7%	22.2%	21.7%
France	10.8%	3.2%	7.0%	18.0%	7.5%
England	7.6%	6.0%	6.8%	14.1%	12.3%
Netherlands	6.0%	6.3%	6.1%	10.8%	12.0%
Brazil	7.4%	4.8%	6.1%	13.7%	11.3%
Argentina	8.6%	1.2%	4.9%	15.5%	3.1%
Belgium	5.7%	3.5%	4.6%	10.9%	7.3%
Portugal	6.6%	2.1%	4.3%	12.7%	4.8%
Uruguay	1.5%	5.0%	3.2%	3.6%	11.0%
Switzerland	2.0%	2.9%	2.5%	5.6%	6.7%
Morocco	3.6%	0.7%	2.1%	7.6%	2.0%
Iran	0.3%	3.2%	1.8%	1.1%	6.5%
USA	1.3%	2.1%	1.7%	3.3%	4.7%
Egypt	0.2%	3.0%	1.6%	0.8%	6.5%
Uzbekistan	0.1%	3.0%	1.6%	0.5%	6.2%
Croatia	2.2%	0.9%	1.5%	4.9%	3.1%
Colombia	1.9%	1.0%	1.5%	4.6%	3.5%
Jordan	0.2%	2.7%	1.4%	0.6%	4.7%
Sweden	0.3%	2.1%	1.2%	1.2%	4.8%
Turkey	1.5%	0.5%	1.0%	3.6%	2.1%
Senegal	1.6%	0.4%	0.9%	4.0%	1.3%
Norway	1.0%	0.9%	0.9%	2.8%	2.7%
Japan	1.3%	0.6%	0.9%	3.5%	1.9%

Table 8: World Cup 2026 Monte Carlo (10,000 sims): championship probabilities for the linear model, the MLP, and their ensemble (average), plus reach-final probabilities; top 24 by ensemble. Shootouts resolved by goalkeeper rating.

4. **MLP: mini-batches, temporal weights, early stopping.** `dixon_coles_loss` takes per-match weights and returns a weighted mean; the `DataLoader` carries each match's temporal-decay weight. Training early-stops on validation NLL, which removed the over-fitting that had pinned the MLP near the base rate.

5. **Leak-safe TRAIN/VAL/TEST split.** Split by match date: TRAIN <2024-01-01, VAL 2024-01-01..2025-10-01 (all selection), TEST ≥2025-10-01 (end 2025 + the World Cup, scored once).

6. **Linear ridge regularization.** Added an $\alpha \sum \beta^2$ penalty on the slope coefficients (not the intercepts or ρ) to `_negative_log_likelihood`; swept five α.

7. **ELO scale / K-factor sensitivity.** `elo_calculator` parameterized by logistic divisor and a fixed K; regenerated match features for divisor $\in \{200, 800, 1600\}$ with $K = 50$.

8. **Reduced feature set & friendly stripping.** Evaluated a 10-feature subset (elo, bench avg, per-match goal difference, attack/defence squad avgs) and a variant with friendlies removed from train+val.

9. **World Cup Monte Carlo.** 10,000-simulation bracket from openfootball/worldcup.json: group stage, best-eight third-place teams slotted into the R32 by a memoized backtracking matching that respects each slot's allowed groups, then the full knockout tree. Matches are neutralized by averaging both home/away orientations.

10. **Goalkeeper-weighted shootouts.** Knockout draws are now decided by $P(A) = \mathrm{gk}_A/(\mathrm{gk}_A + \mathrm{gk}_B)$ using a per-squad `gk_rating` (best goalkeeper overall, falling back to defence average). Previously draws were resolved by overall strength λ.

	Linear			MLP		
Team	GK	λ	Δ	GK	λ	Δ
Spain	14.0%	15.9%	-2.0	15.4%	20.6%	-5.2
France	10.8%	12.1%	-1.3	3.2%	2.7%	+0.5
Argentina	8.6%	9.1%	-0.5	1.2%	0.6%	+0.6
England	7.6%	8.3%	-0.7	6.0%	5.7%	+0.3
Germany	7.6%	7.9%	-0.3	22.4%	26.9%	-4.5
Brazil	7.4%	7.8%	-0.4	4.8%	3.9%	+0.9
Portugal	6.6%	6.9%	-0.3	2.1%	1.2%	+0.9
Netherlands	6.0%	5.9%	+0.1	6.3%	5.5%	+0.8
Belgium	5.7%	5.1%	+0.6	3.5%	2.7%	+0.8
Morocco	3.6%	3.2%	+0.4	0.7%	0.4%	+0.3
Croatia	2.2%	1.9%	+0.3	0.9%	0.4%	+0.5
Switzerland	2.0%	1.9%	+0.1	2.9%	2.0%	+0.9

Table 9: Shootout-resolution sensitivity: championship probability when knockout draws are decided by goalkeeper rating (GK) vs. overall strength (λ). Δ in percentage points.

11. **Thin-squad repair (FC26 data gaps).** Some nations have very small FC26 squads (Egypt 10 players, Iran 6, Uzbekistan 5), giving a 0 bench average and an inflated dispersion that the MLP extrapolated to absurd odds (Egypt champion 65%). Fixes in `squad_features.py`: (i) std features converted to standard error (std/$\sqrt{n}$) and capped at the 85th percentile; (ii) bench average imputed for squads $\leq$20 as squad_avg minus the mean squad-to-bench gap of full squads; (iii) `gk_rating` extracted, with a defence-average fallback.